# Annotating Solanus Casey's writings once for every downstream use

**A single, well-designed annotation pass across the ~1,000 pages of Fr. Solanus Casey's collected writings can serve all four pipelines—style fine-tuning, Azure AI Search, knowledge graph construction, and RAG—if the schema captures both physical layout zones and semantic content types as independent facets.** The key insight is that each pipeline consumes different slices of the same annotation data: fine-tuning needs only authored text filtered by region type; search needs chunked content with rich facetable metadata; the knowledge graph needs entity spans with canonical identifiers; and RAG needs everything tied together with reading order and provenance. This report delivers the complete annotation architecture: a concrete label taxonomy, a production-ready JSON schema, pipeline-specific export strategies, and a step-by-step workflow for annotating every page exactly once.

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## The Solanus Casey corpus is richer and stranger than most assume

The four volumes plus three appendices published at solanuscasey.org in 2023 contain a remarkably heterogeneous collection. The backbone is **seven ledger-style notebooks** spanning 1923–1956, Zieglers with over **6,000 terse entries** documenting prayer requests, Seraphic Mass Association enrollments, and reported healings Zieglers ("Patrick McCue enrolled for one year Feb. 9, by fellow motorman—reported today entirely cured of both gangrene and diabetes. Doctors baffled!"). CapDox Solanuscasey Alongside these sit hundreds of **personal letters** to family (especially his sister Margaret LeDoux Solanus Center and brother Msgr. Edward Casey), Solanuscasey friends, and spiritual directees. Detroit Catholic The corpus also includes **novitiate spiritual resolutions** structured as numbered lists, **biblical reflections** on themes like gratitude, patience, and the Eucharist, CapDox capuchin **copied and adapted prayers**, Blogger occasional **poetry and verse**, **marginal inscriptions on photographs**, Solanuscasey and miscellaneous **note cards and leaflets** he enclosed with letters.

His writing style is distinctive and presents specific challenges for annotation. Nearly every letter opens or closes with formulaic pious exclamations—**"Praised Be Jesus Christ!"**, **"Thanks Be to God!"**, Solanuscasey **"God Bless You and Yours"**— ThriftBooks Solanuscasey and his signature refrain **"Thank God ahead of time"** recurs throughout. Good Catholic Wordpress The prose is warm, conversational, and deceptively simple, with what biographers call "childlike simplicity" masking theological depth. Catholic Exchange capuchin His spelling is **highly inconsistent** Solanuscasey ("temporl," "highten," Solanuscasey "enemia/enimic/inemic" for anemia/anemic), reflecting his well-documented academic struggles that led to his ordination as a "simplex priest." Wikipedia The Tablet This idiosyncratic orthography must be preserved verbatim for style mimicry and is a known challenge for search indexing.

The physical documents were originally handwritten, later transcribed by **Brother Leo Wollenweber, OFM Cap.**, Solanuscasey who catalogued "eight drawers full of favors and notebooks" at the Solanus Casey Center. Detroit Catholic The PDFs available online appear to be scans of these typewritten transcriptions, notarized by Rev. Charles Dillen as part of the canonical beatification process. Solanuscasey Each page typically contains archivist headers ("this document is a faithful reproduction..."), catalog markings, and sometimes editorial notes alongside the original content—exactly the mixed-format challenge this annotation system must solve.

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## A two-facet label taxonomy separates what you see from what it means

The central design decision is to maintain **two independent label dimensions**: structural labels describing the physical layout zone, and semantic labels describing the content's meaning. This follows the SegmOnto controlled vocabulary for historical documents Episciences and the faceted annotation approach validated in CHI 2023 research, which showed hierarchical labeling schemes improve annotator performance by up to +0.40 F1 ACM Digital Library on difficult examples compared to flat schemes.

### Structural labels (physical zones)

These describe *what kind of region this is on the page*, regardless of content:

```
STRUCTURAL_LABELS
├── primary_content
|   ├── main_body             # Core authored text block
|   ├── salutation            # Opening greeting ("My Dear Sister M...")
|   ├── closing               # Closing formula ("Yours in Christ...")
|   ├── signature             # Signature block
|   ├── postscript            # P.S. additions after closing
|   ├── date_line             # Date written on document
|   ├── address_block         # Recipient/sender address
|   └── continuation          # Content continuing from previous page
├── secondary_content
|   ├── margin_note_author    # Solanus's own marginal additions
|   ├── interlinear_note      # Text inserted between lines
|   ├── footnote              # Notes at bottom of text area
|   └── insertion             # Text inserted with carets/arrows
├── archivist_layer
|   ├── archivist_header      # "This document is a faithful reproduction..."
|   ├── archivist_annotation  # Editorial notes about the document
|   ├── catalog_marking       # Reference numbers, shelf marks
|   ├── draft_indicator       # Notes about draft/unfinished status
|   ├── image_description     # Archivist descriptions of images on cards
|   ├── cross_reference       # References to other documents in collection
|   └── stamp                 # Library/archival stamps
├── paratext
|   ├── page_number_printed   # Printed page numbers (from PDF)
|   ├── page_number_original  # Original notebook page numbers
|   ├── header                # Running headers
|   └── footer                # Running footers
├── non_text
|   ├── illustration          # Drawings, holy cards, photographs
|   ├── decoration            # Decorative elements
|   └── blank_space           # Intentional blank areas
└── damage
    ├── illegible             # Text present but unreadable
    ├── torn_area             # Physical damage
    └── faded                 # Partially legible deteriorated text
```

### Semantic labels (content meaning)

These describe *what the content is about*, applied independently:

```
SEMANTIC_LABELS
├── theological
|   ├── scripture_quotation   # Direct Bible quote
|   ├── scripture_reference   # Citation without full text
|   ├── prayer_text           # Prayer (original or copied)
|   ├── doctrinal_teaching    # Theological doctrine
|   ├── exegesis              # Biblical interpretation
|   └── liturgical            # Liturgical/sacramental content
├── pastoral
|   ├── spiritual_counsel     # Advice/guidance to others
|   ├── personal_reflection   # Internal thoughts/feelings
|   ├── gratitude_expression  # Thanksgiving (his dominant theme)
|   ├── suffering_reflection  # On redemptive suffering
|   └── providence_trust      # On confidence in God
├── documentary
|   ├── favor_report          # Documented healing/favor
|   ├── enrollment_record     # SMA enrollment notation
|   ├── news_report           # Reporting events
|   ├── administrative        # Organizational business
|   └── biographical_detail   # About specific persons
├── literary
|   ├── poetry                # Verse/poetry
|   ├── hymn                  # Hymn text
|   └── copied_text           # Text copied from another source
└── correspondence
    ├── personal              # Personal/family matters
    ├── request               # Asking for something specific
    └── greeting_formula      # Formulaic opening/closing phrases
```

**The critical advantage of two independent facets** is that dual-purpose content resolves naturally. A salutation containing theological content gets `structural_label: "salutation"` plus `semantic_labels: ["greeting_formula", "prayer_text"]`. A notebook entry recording a healing gets `structural_label: "main_body"` plus `semantic_labels: ["favor_report", "gratitude_expression"]`. No information is lost, and each pipeline extracts what it needs: fine-tuning filters on structural labels to get authored text; the knowledge graph filters on semantic labels to find entity-rich theological content; search indexes both dimensions as independent facets.

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## The JSON schema must serve as a single source of truth for all pipelines

The following schema captures everything needed for all four downstream uses. It uses **COCO-style flat polygon coordinates** (the most widely supported format for ML training) CVAT alongside **Dublin Core-aligned metadata** (the standard for archival interoperability) and **inline entity spans** (for knowledge graph construction). Each page produces one JSON document; pages belonging to the same logical document share a `document_id`.

```json
{
  "schema_version": "1.0.0",
  "page_id": "vol4-p0042",
  "source_file": "Volume-4.pdf",
  "source_page_index": 42,

  "image": {
    "file_name": "vol4_page_042.png",
    "width": 2550,
    "height": 3300,
    "dpi": 300,
    "format": "image/png"
  },

  "documents_on_page": [
    {
      "document_id": "vol4-letter-037",
      "document_type": "letter",
      "document_subtype": "personal_correspondence",
      "title": "Letter to Margaret LeDoux",
      "date_composed": "1943-07-15",
      "date_precision": "exact",
      "date_display": "July 15, 1943",
      "author": "Solanus Casey",
      "recipient": "Margaret Casey LeDoux",
      "recipient_role": "sister",
      "origin_location": "St. Bonaventure Monastery, Detroit",
      "language": ["en"],
      "topics": ["gratitude", "family_news", "divine_providence"],
      "biblical_references": [
        {"book": "Romans", "chapter": 8, "verse": "28"}
      ],
      "people_mentioned": ["Margaret Casey LeDoux", "Edward Casey"],
      "places_mentioned": ["Detroit", "Portland"],
      "physical_condition": "good",
      "completeness": "complete",
      "is_draft": false,
      "page_range_in_document": {"start": 1, "end": 2},
      "this_page_position": 1,
      "total_pages_in_document": 2,
      "collection": "Collected Writings, Volume 4",
      "archive_reference": "SCC-CW-V4-037"
    }
  ],

  "annotations": [
    {
```

```
    "id": "vol4-p042-r001",
    "document_id": "vol4-letter-037",
    "structural_label": "archivist_header",
    "semantic_labels": [],
    "polygon": [50, 30, 2500, 30, 2500, 120, 50, 120],
    "bbox": [50, 30, 2450, 90],
    "reading_order": null,
    "transcription": "This document is a faithful reproduction of the original hel
    "hand": "archivist",
    "include_in_finetuning": false,
    "include_in_search": false,
    "include_in_rag": false,
    "flags": [],
    "entities": [],
    "notes": ""
  },
  {
    "id": "vol4-p042-r002",
    "document_id": "vol4-letter-037",
    "structural_label": "date_line",
    "semantic_labels": [],
    "polygon": [1800, 180, 2400, 180, 2400, 220, 1800, 220],
    "bbox": [1800, 180, 600, 40],
    "reading_order": 1,
    "transcription": "Detroit, July 15, 1943",
    "hand": "solanus",
    "include_in_finetuning": true,
    "include_in_search": true,
    "include_in_rag": true,
    "flags": [],
    "entities": [
      {
        "text": "Detroit",
        "type": "place",
        "normalized": "Detroit, Michigan",
        "char_start": 0,
        "char_end": 7
      },
      {
        "text": "July 15, 1943",
        "type": "date",
        "normalized": "1943-07-15",
        "char_start": 9,
        "char_end": 22
      }
    ],
    "notes": ""
  },
  {
    "id": "vol4-p042-r003",
```

```
    "document_id": "vol4-letter-037",
    "structural_label": "salutation",
    "semantic_labels": ["greeting_formula", "prayer_text"],
    "polygon": [150, 260, 1200, 260, 1200, 340, 150, 340],
    "bbox": [150, 260, 1050, 80],
    "reading_order": 2,
    "transcription": "Praised Be Jesus Christ!\nMy Dear Sister Margaret,",
    "hand": "solanus",
    "include_in_finetuning": true,
    "include_in_search": true,
    "include_in_rag": true,
    "flags": [],
    "entities": [
      {
        "text": "Margaret",
        "type": "person",
        "normalized": "Margaret Casey LeDoux",
        "char_start": 38,
        "char_end": 46
      }
    ],
    "notes": ""
  },
  {
    "id": "vol4-p042-r004",
    "document_id": "vol4-letter-037",
    "structural_label": "main_body",
    "semantic_labels": ["personal_reflection", "gratitude_expression", "divine_pro
    "polygon": [150, 360, 2400, 360, 2400, 2800, 150, 2800],
    "bbox": [150, 360, 2250, 2440],
    "reading_order": 3,
    "transcription": "Thanks Be to God for the wonderful news about...",
    "transcription_confidence": 0.95,
    "hand": "solanus",
    "include_in_finetuning": true,
    "include_in_search": true,
    "include_in_rag": true,
    "flags": [],
    "entities": [
      {
        "text": "Edward",
        "type": "person",
        "normalized": "Msgr. Edward Casey",
        "char_start": 52,
        "char_end": 58
      }
    ],
    "notes": ""
  },
```

```
    {
      "id": "vol4-p042-r005",
      "document_id": "vol4-letter-037",
      "structural_label": "page_number_original",
      "semantic_labels": [],
      "polygon": [1250, 3200, 1350, 3200, 1350, 3250, 1250, 3250],
      "bbox": [1250, 3200, 100, 50],
      "reading_order": null,
      "transcription": "237",
      "hand": "archivist",
      "include_in_finetuning": false,
      "include_in_search": false,
      "include_in_rag": false,
      "flags": [],
      "entities": [],
      "notes": ""
    }
  ],

  "page_flags": [],

  "annotation_metadata": {
    "annotator_id": "researcher-01",
    "annotation_date": "2026-03-25T14:30:00Z",
    "tool_version": "1.2.0",
    "time_spent_seconds": 840,
    "review_status": "submitted"
  }
}
```


**Three boolean fields—`include_in_finetuning`, `include_in_search`, `include_in_rag`—are the most operationally important fields in the schema.** They let each pipeline's export script filter regions without re-interpreting structural labels every time. Set them during annotation based on simple rules: archivist content gets all three set to `false`; authored content gets all three set to `true`; page numbers and catalog markings get all `false`. This eliminates ambiguity in downstream processing.

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## Each pipeline consumes different slices through transformation scripts

The annotation schema acts as a universal source; each pipeline applies filters and reshaping to extract what it needs.

### Fine-tuning for Solanus's spiritual voice

The fine-tuning pipeline extracts only regions where `include_in_finetuning == true` and `hand == "solanus"`, concatenates them in `reading_order`, and structures them as chat-format training examples. For ~1,000 pages with an average of 200–400 words of authored content per page, expect **3,000–8,000 training examples** after chunking at natural document boundaries (complete letters, individual notebook entries, complete prayers).

The recommended format for LoRA/QLoRA fine-tuning on a 7B–13B parameter base model:

```json
{"messages": [
  {"role": "system", "content": "You are Blessed Solanus Casey, a Capuchin friar kno
  {"role": "user", "content": "Write a letter offering spiritual encouragement about
  {"role": "assistant", "content": "Praised Be Jesus Christ!\nMy Dear Friend in Chri
]}
```

**Critical: preserve original spelling, punctuation, and idiosyncratic orthography.** Solanus's misspellings, creative punctuation, and phonetic renderings are essential style signals. Do not normalize "temporl" to "temporal" or "highten" to "heighten"—these are features, not bugs, for style mimicry. Use LoRA with rank **r=8–16**, learning rate **1e-4 to 2e-5**, and train for **3–5 epochs**. The small corpus makes parameter-efficient fine-tuning essential to avoid catastrophic forgetting of the base model's language abilities.

Exclude all regions with `hand != "solanus"`, all archivist headers, all catalog markings, and all editorial notes. Include salutations and closings—his formulaic "Praised Be Jesus Christ!" and "God Bless You and Yours" are among his most distinctive stylistic features Good Catholic and must be learned by the model.

### Azure AI Search with full feature stack

For search indexing, export all regions where `include_in_search == true`, grouped by `document_id`. Each logical document becomes a parent record; paragraphs within become child chunks of **512 tokens with 25% overlap** (Microsoft's recommended starting point). The index schema should include:

- `content` (Edm.String, searchable/retrievable): concatenated authored text
- `content_vector` (Collection(Edm.Single), searchable): embedding from text-embedding-3-large
- `document_type` (Edm.String, filterable/facetable): letter, notebook_entry, prayer, poem, spiritual_resolution
- `date_composed` (Edm.DateTimeOffset, filterable/sortable): ISO date
- `recipient` (Edm.String, filterable/facetable): canonical recipient name
- `topics` (Collection(Edm.String), filterable/facetable): semantic labels from annotation
- `people_mentioned` (Collection(Edm.String), filterable/facetable): extracted from entity spans
- `places_mentioned` (Collection(Edm.String), filterable/facetable): extracted from entity spans
- `biblical_references` (Collection(Edm.String), filterable/facetable): scripture citations

Configure skillsets to run **OCR** (if working from scans), **Entity Recognition**, **Key Phrase Extraction**, and **Azure OpenAI Embedding** skills. PyPI The knowledge store should project three types: **table projections** for entities and key phrases (queryable via Power BI), **object projections** for full enriched documents, and **file projections** for page images. Microsoft Learn Semantic ranking should prioritize `content` as the primary content field and `topics` plus extracted key phrases as keyword fields—keep metadata IDs and dates out of semantic ranking fields, as they add noise. OneUptime

### Knowledge graph construction

The entity spans tagged during annotation seed the knowledge graph directly. Extract these node types from the annotations:

**Person** nodes (Solanus, recipients, family members, fellow friars, people requesting favors), **Place** nodes (Detroit, St. Bonaventure Monastery, Huntington IN, Portland OR), **Organization** nodes (Capuchin Order, Seraphic Mass Association, specific parishes), **Event** nodes (reported healings, ordination, assignments to new posts), **TheologicalConcept** nodes (gratitude, divine providence, redemptive suffering, the Eucharist), **BiblicalReference** nodes (specific scripture citations), and **Document** nodes (the source documents themselves as first-class entities).

Key relationships include `WROTE_TO` (Person→Person), `MENTIONS` (Document→Entity), `DISCUSSES` (Document→TheologicalConcept), `REFERENCES` (Document→BiblicalReference), `LOCATED_AT` (Person→Place at a given time), `RESPONDS_TO` (Document→Document for correspondence chains), and `MEMBER_OF` (Person→Organization). Each relationship edge should carry `source_document_id`, `page_id`, and `region_id` for full provenance back to the annotation.

Maintain a **canonical entity registry** during annotation—a simple lookup table mapping variant spellings ("Fr. Edward," "Edward Casey," "Msgr. Edward Casey," "my brother Edward") to canonical identifiers. The `normalized` field in each entity span serves this purpose. This registry is the single most valuable artifact for knowledge graph quality, and building it incrementally during the one-pass annotation is far more efficient than post-hoc entity resolution.

### RAG and chatbot Q&A

For RAG, use **hierarchical chunking**: full documents as Level 1 (for broad context retrieval) and paragraph-level chunks of **400–512 tokens** as Level 2 (for precise answer extraction). Every chunk carries the full metadata payload from its parent document annotation—`document_type`, `date_composed`, `recipient`, `topics`, `people_mentioned`, `places_mentioned`. Prefix each chunk with contextual framing:

- Letters: `"[Letter to Margaret Casey LeDoux, July 15, 1943, from Detroit] ..."`
- Notebook entries: `"[Notebook entry, November 8, 1923 — Seraphic Mass Association favor report] ..."`
- Prayers: `"[Prayer, adapted from Phillips Brooks] ..."`

Use **hybrid search** (BM25 keyword + vector similarity) with semantic re-ranking. DEV Community Solanus's archaic and idiosyncratic vocabulary may not embed well with modern models—BM25 catches exact keyword matches that vector search misses, while vectors catch semantic intent. Retrieve **5–8 chunks** per query (2,000–4,000 tokens of context). Always return `document_id`, `page_id`, `date_composed`, and `document_type` with each answer for citation and verifiability.

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## The one-pass annotation workflow processes each page in a fixed seven-step sequence

The workflow is designed around a **15–25 minute cycle per page** (excluding transcription, which varies by content density). For ~1,000 pages, this translates to roughly 250–400 hours of annotation work. The fixed sequence ensures completeness and prevents the annotator from forgetting steps.

**Phase 0 — Setup (before any annotation begins).** Pilot-annotate 25–30 pages selected to represent every document type (letters from different periods, notebook entries, prayers, poetry, pages with archivist annotations, damaged pages). Use these to calibrate the label taxonomy, refine edge case decisions, and establish timing benchmarks. Create a living annotation guideline document with visual examples of correct labeling for each structural and semantic type. Build the initial canonical entity registry from the index already published at solanuscasey.org (which cross-references all four volumes). Solanuscasey Solanuscasey

**Phase 1 — Per-page annotation in fixed order:**

1. **Page metadata** (1–2 min): Record source file, page index, assign to `document_id`. If this page begins a new document, create a new document record with type, date, recipient, etc. If continuing an existing document, link to the existing `document_id`.
2. **Structural segmentation** (4–6 min): Draw polygons around every distinct content zone, working top-to-bottom, left-to-right. Assign one structural label per polygon. This is the most time-intensive step and should be done first while attention is freshest.
3. **Reading order** (1 min): Number authored-content regions in the order they should be read. Skip archivist headers, page numbers, and stamps (set reading_order to `null`).
4. **Semantic tagging** (2–4 min): Apply one or more semantic labels to each authored-content region. Archivist regions get no semantic labels (empty array).
5. **Entity tagging** (2–3 min): Within each region's transcription, mark person names, places, dates, scripture references, and organizations with character offsets and canonical forms. Consult the entity registry; add new entries as they appear.
6. **Pipeline inclusion flags** (30 sec): Set `include_in_finetuning`, `include_in_search`, `include_in_rag` booleans. Default rules: Solanus-authored content = all true; archivist content = all false; page numbers/stamps = all false.
7. **Flags and self-review** (1 min): Apply any edge-case flags (`illegible`, `damaged`, `needs_review`, `uncertain_label`). Quick visual scan of all regions for completeness.

**Phase 2 — Automated validation (runs after each page save):**

- Every page must have ⩾1 annotation region
- Every region must have exactly one structural label from the controlled vocabulary
- Reading order values must be sequential with no gaps within a document
- Required document-level metadata fields must be populated (`document_type`, `date_composed` or `date_precision: "unknown"`, `author`)
- No two regions of the same structural type should overlap by more than 10% area (catches accidental duplicates)
- Pages flagged `needs_review` are queued for expert review

**Phase 3 — Periodic quality checks (every 50–100 pages):**

Review a random 10–15% sample. Track **completeness** (percentage of page area covered by annotations, target >95%), **label consistency** (same content type labeled the same way across pages), and **throughput** (pages per hour, to predict project timeline). Update annotation guidelines with new edge-case decisions.

**Edge case protocols**

**Multi-document pages** (one physical page contains the end of one letter and the start of another): Draw separate polygon regions for each document's content. Assign different `document_id` values to each region. The `documents_on_page` array in the JSON lists both documents. Use the `continuation` structural label for content that began on a previous page.

**Verso/back-of-document notes**: Treat as a separate region with `structural_label: "margin_note_author"` if by Solanus, or `archivist_annotation` if by an archivist. Link to the recto document via `document_id`.

**Illegible text**: Draw the polygon, assign the best-guess structural label based on position and context, flag as `illegible`, transcribe whatever is visible with `[illegible]` markers for unreadable portions. Set `transcription_confidence` to reflect legibility.

**Mixed languages** (Solanus occasionally used Latin phrases): Keep in a single region if interleaved with English. Set the region's `language` field to the primary language and flag `mixed_language` if significant. For fine-tuning, preserve the multilingual content—it's part of his authentic voice.

**Crossed-out or revised text**: Draw the polygon, label with the original structural type, flag `crossed_out`, and transcribe the original text. Add a note explaining the revision if a replacement is visible.

**Inserted clippings or holy cards**: Treat as `illustration` structural type with `image_description` if the archivist described the image, or as a separate document if it contains authored text.

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## How to organize the output for folder-based grouping

The `document_id` field enables automatic grouping into the researcher's desired folder structure. A simple script can read the JSON annotations and organize exported files as:

```
Volume-4/
├── letters/
│   ├── to_margaret_ledoux/
│   │   ├── letter_037/
│   │   │   ├── page_042.json
│   │   │   ├── page_043.json
│   │   │   └── document_metadata.json
│   │   └── letter_041/
│   │       └── ...
│   └── to_edward_casey/
│       └── ...
├── notebook_entries/
│   ├── notebook_1/
│   │   ├── entry_001/
│   │   └── ...
│   └── notebook_7/
├── prayers/
├── poetry/
└── spiritual_resolutions/
```

The `document_metadata.json` at each document folder level aggregates all page-level annotations into a single document record, concatenating transcriptions in reading order and merging entity lists with deduplication. This aggregated file is what the Azure AI Search indexer, knowledge graph builder, and RAG chunker actually consume.

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## Conclusion: design for the export, not the annotation

The most important takeaway is architectural: **the annotation schema should capture maximum information at the region level, and each pipeline's ingestion script should filter and reshape that information according to its needs.** The three boolean inclusion flags (`include_in_finetuning`, `include_in_search`, `include_in_rag`) prevent downstream ambiguity. The two-facet label taxonomy (structural × semantic) prevents information loss from dual-purpose content. The inline entity spans with canonical identifiers eliminate the need for a separate NER pass. And the fixed seven-step annotation sequence ensures every page is complete after a single pass.

For the Solanus Casey corpus specifically, the **canonical entity registry** deserves investment disproportionate to its apparent simplicity. His notebooks reference thousands of people by inconsistent name variants across 30 years of entries— Zieglers Wordpress resolving "Mrs. B." on page 400 to the same person as "Bernadette Sullivan" on page 12 is the difference between a useful knowledge graph and a disconnected one. Build the registry incrementally during annotation using the published index as a seed, and it will pay dividends across every pipeline.

The resulting annotation dataset—roughly 1,000 JSON files with polygon regions, dual-facet labels, entity spans, and rich metadata—becomes a **permanent scholarly resource** that can support not just the four planned pipelines but future uses not yet imagined: computational stylometry, network analysis of his correspondence community, temporal analysis of his theological development, or training layout analysis models for similar Capuchin archival collections.